```
import pandas as pd
import numpy as np

# Visualization
import matplotlib.pyplot as plt

# Sklearn
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_sc
```

```
df = pd.read_csv('/content/insurance.csv')
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

```
df_encoded = pd.get_dummies(df, drop_first=True)
df_encoded.head()
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest
0	19	27.900	0	16884.92400	False	True	False
1	18	33.770	1	1725.55230	True	False	False
2	28	33.000	3	4449.46200	True	False	False
3	33	22.705	0	21984.47061	True	False	True
4	32	28.880	0	3866.85520	True	False	True

Next steps: Generate code with df_encoded New interactive sheet

```
X = df_encoded.drop('charges', axis=1)
y = df_encoded['charges']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("MSE:", mse)
print("R2 Score:", r2)
```

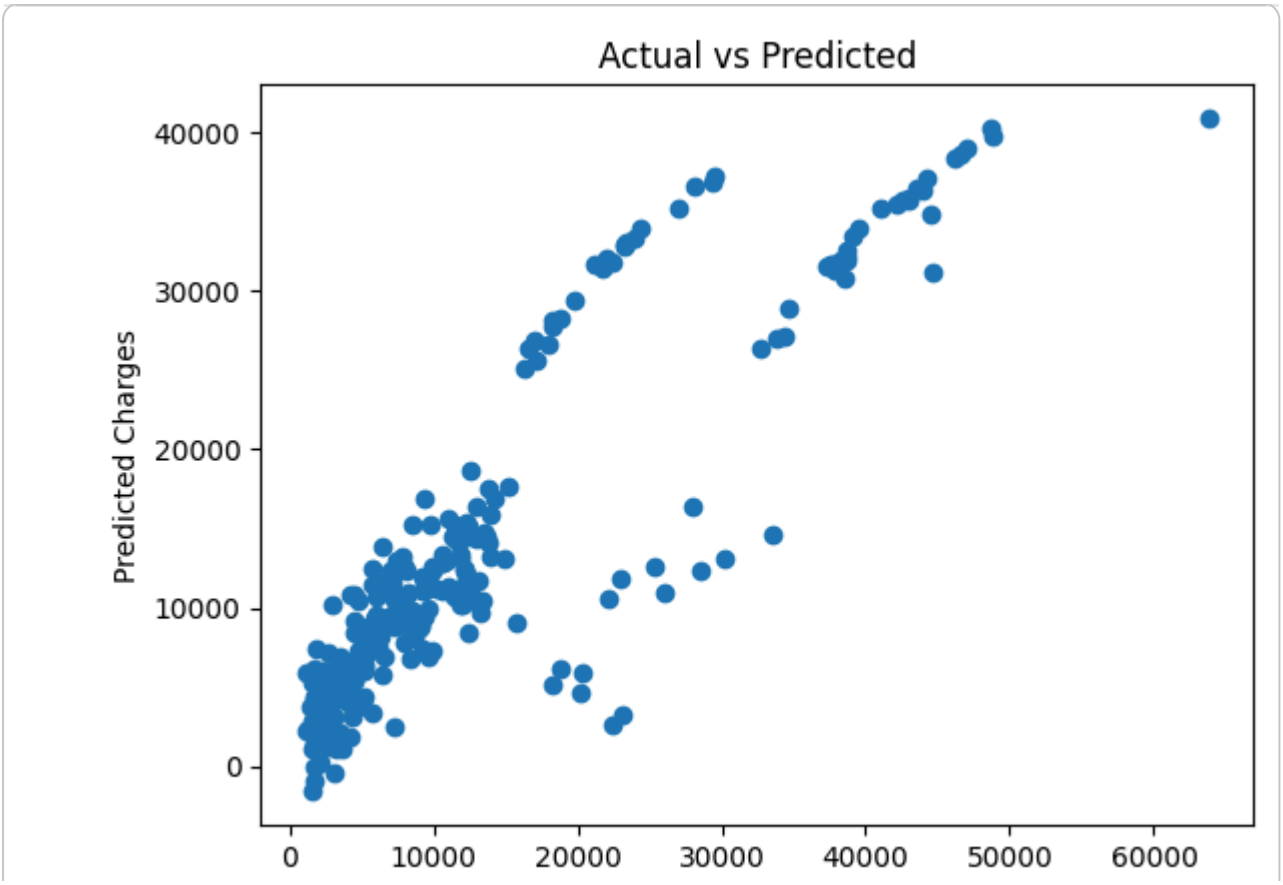
```
MAE: 4181.19447375365
MSE: 33596915.85136145
R2 Score: 0.7835929767120724
```

```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
})

coefficients.sort_values(by='Coefficient', ascending=False)
```

	Feature	Coefficient
4	smoker_yes	23651.128856
2	children	425.278784
1	bmi	337.092552
0	age	256.975706
3	sex_male	-18.591692
5	region_northwest	-370.677326
6	region_southeast	-657.864297
7	region_southwest	-809.799354

```
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Actual vs Predicted")
plt.show()
```


Actual vs Predicted
Predicted Charges
40000
30000
20000
10000
0
0
10000
20000
30000
40000
50000
60000